

Counting and Probability

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Preface

Placeholder

1 Some Counting Problems

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Answers/Solutions

2 Pigeonhole Principle

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3 Division Principle

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3.1 A problem about cookies

3.2 The Division Principle

3.3 More Counting Problems

4 Counting Problems

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4.1 Donut problems

4.2 More Problems

5 Probability

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5.1 Notation Note

5.2 Probability Axioms

5.3 Some Additional Probability Rules

5.4 Exercises

6 Conditional Probability

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6.1 Exercises

6.2 Product Rule

6.3 Independence

6.4 Exercises

7 Probability Trees and Bayes

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7.1 Bob's Beautiful Boxes

7.2 Breast cancer screening

7.3 Disease Testing

7.4 Random Primality Testing

7.5 Hashing

8 Random Variables

Placeholder

8.1 Describing Random Variables

8.2 Probability Tables

First example

Creating a probability table

8.3 Probability Functions

8.4 Binomial Distributions

9 Expected Value

Placeholder

9.1 An Example: GPA

9.2 Generalizing